

Virtual LAN (VLAN)

A Virtual local area network is a logical way to split a single physical network into multiple separate virtual networks.

- It reduces unnecessary broadcast traffic, enhances security, and improves network performance, flexibility, and manageability.
- Traditionally, broadcast domains were separated using Layer 3 devices (routers).
- **Same VLAN:** Devices communicate directly within the same broadcast domain.
- **Different VLANs:** Communication requires Inter-VLAN Routing via a router or a Layer 3 switch (SVI).

VLAN ID Ranges

- **VLAN 0 and VLAN 4095:** Reserved VLAN IDs are defined by IEEE 802.1Q and used internally for protocol operations and cannot be configured or assigned to ports.
- **VLAN 1 (Default VLAN):** The default VLAN on Cisco switches and all switch ports are assigned to VLAN 1 by default. It cannot be deleted and is commonly used for control and management protocols such as STP, CDP, and VTP.
- **VLAN 2 to VLAN 1001 (Normal VLAN Range):** This range is used for regular VLAN configuration. VLANs in this range are fully configurable, editable, and deletable, and are stored in the switch's VLAN database.
- **VLAN 1002 to VLAN 1005:** Reserved VLANs for legacy network technologies such as FDDI and Token Ring. These VLANs are predefined and cannot be removed, even if the technologies are no longer in use.
- **VLAN 1006 to VLAN 4094 (Extended VLAN Range):** Designed for large-scale networks requiring a high number of VLANs. VLANs in this range are stored in the **running configuration** and typically require the switch to operate in VTP transparent mode.

Types of VLAN Links

- **Access Link:** Connects a VLAN-unaware end device (such as a PC) to a VLAN-aware switch. Frames transmitted over an access link are untagged and belong to a single VLAN.

enable

configure terminal

vlan 10

name Sales

exit

- **Trunk Link:** Connects VLAN-aware devices such as switch-to-switch or switch-to-router links. It carries traffic from multiple VLANs using IEEE 802.1Q tagging.

configure terminal

interface GigabitEthernet 0/1

switchport mode trunk

#Specify Allowed VLANs

switchport trunk allowed vlan 10,20,30

end

- **Hybrid Link:** Supports both tagged and untagged frames on the same link. It is typically vendor-specific and not part of the IEEE 802.1Q standard.

Advantages

- **Improved Security:** Sensitive traffic is isolated within separate VLANs.
- **Enhanced Performance:** Limits broadcast and multicast traffic, reducing unnecessary network load.
- **Simplified Management:** Logical grouping of users or departments (e.g., HR, Finance) simplifies administration.
- **Flexibility:** Devices can be moved between VLANs without physical cable changes.
- **Cost Efficiency:** Reduces the need for multiple physical networks and excessive routing hardware.
- **Scalability:** Large networks can be divided into manageable segments.

Disadvantages

- **Increased Configuration Complexity:** Requires careful planning and configuration.
- **VLAN ID Limitations:** Standard VLAN IDs are limited, which can affect very large networks.
- **Security Risks:** Misconfigurations may allow VLAN hopping attacks.
- **Interoperability Issues:** Compatibility problems may arise with non-standard or legacy devices.
- **Troubleshooting Difficulty:** Isolated traffic flows can make fault diagnosis more complex.

VLAN Hopping

VLAN Hopping is an attack where the attacker is able to send traffic from one VLAN into another.

IEEE 802.1Q

IEEE 802.1Q is the networking standard that supports Virtual LANs (VLANs) on an Ethernet network. When you have a trunk link connecting two switches, or a switch and a router, that single physical connection needs a way to carry traffic for multiple VLANs while keeping them logically separated.

How 802.1Q Works: The 4-Byte Tag

802.1Q accomplishes VLAN separation by inserting a 4-byte tag into the original Ethernet frame header. This tag is inserted into the frame right after the Source MAC Address and before the original EtherType/Length field.

Because this tag is injected into the middle of an existing frame, the frame's overall size increases, and its contents change. Consequently, the switch must recalculate the Frame Check Sequence (FCS) trailer at the end of the frame so the receiving device doesn't drop it as a corrupted packet.

Native VLAN

The major exception to this tagging process is the **Native VLAN**.

- By default, the Native VLAN is VLAN 1.
- Traffic belonging to the Native VLAN is sent completely **untagged** across the trunk link. The switch does not insert the 4-byte 802.1Q header into these frames.
- When a receiving switch gets an untagged frame on an 802.1Q trunk port, it automatically forwards that frame into whatever VLAN is configured as the Native VLAN on that specific interface.

```
configure terminal
interface GigabitEthernet 0/1
switchport trunk native vlan 99
end
```

End Devices (Access Ports)

Standard network devices—like desktop computers, laptops, and typical printers—do not understand VLAN tags.

- When a PC sends data, it sends **untagged traffic**.
- The switch port connected to that PC is configured as an "access port" assigned to a specific VLAN (e.g., VLAN 10).
- When the switch receives the untagged frame from the PC, the switch internally tags it with VLAN 10 so it knows how to route it through the network.
- Before the switch sends a frame back out to the destination PC, it strips the tag off, delivering an **untagged frame** to the device.

Switch-to-Switch (Trunk Ports and the Native VLAN)

When traffic travels between two switches over a trunk link, it usually carries data for multiple VLANs. The switch must add a 4-byte 802.1Q tag to the frames so the receiving switch knows which VLAN the traffic belongs to.

However, there is one exception on a trunk link: the **Native VLAN**.

- You configure one VLAN on the trunk to be the Native VLAN (default is VLAN 1).
- When a switch needs to send traffic that belongs to the Native VLAN across the trunk, it deliberately **does not add a tag**. It sends it as **untagged traffic**.
- When the receiving switch gets an untagged frame on its trunk port, it assumes that frame belongs to its locally configured Native VLAN and processes it accordingly.

Note:

Untagged Traffic: A normal network frame without a VLAN ID.

Untagged VLAN (Native VLAN): The specific VLAN on a trunk link where the switches agree to send traffic without a tag.

Troubleshooting Command list

General VLAN Commands

- show vlan brief
- show vlan id <vlan-id>
- show mac address-table vlan <vlan-id>
- show spanning-tree vlan <vlan-id>
-

Access Port Commands

- show interfaces status
- show interfaces <interface-id> switchport
- show mac address-table interface <interface-id>
- show running-config interface <interface-id>

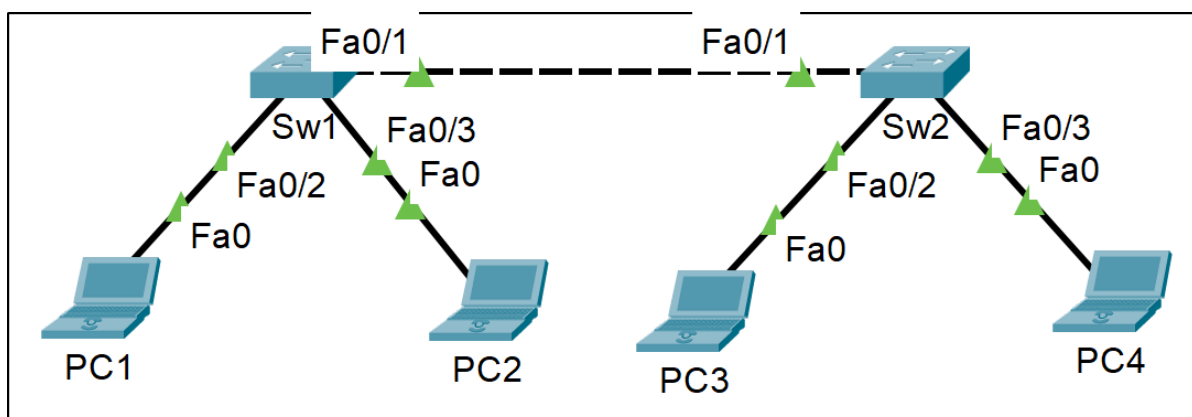
Trunk Port Commands

- show interfaces trunk
- show interfaces <interface-id> trunk
- show interfaces <interface-id> switchport
- show cdp neighbors
- show cdp neighbors detail

General Commands

- show logging
- show errdisable recovery
- show interfaces <interface-id> counters errors

Lab 2 – VLAN and Trunking



(Above design is built on the Cisco Packet Tracer simulator. Build the same diagram with the port details and follow the task checklists as below.)

Tasks:

- Set the hostnames on Sw1 and Sw2 as given in the diagram.
- Configure VLAN 10 AND 20 on Sw1 and Sw2. Enable static trunking between two switches, only allow VLAN 10 and 20 through the trunk port.
- Configure access ports on Sw1 and Sw2, PC1 and PC3 will be part of VLAN 10 while the PC2 and PC4 should be part of VLAN 20.
- Configure IP addresses on the hosts as given below.
 - o PC1 : 192.168.1.1/27
 - o PC2 : 192.168.1.42/27
 - o PC3 : 192.168.1.3/27
 - o PC4 : 192.168.1.44/27
- Once configured, hosts within same VLAN should be able to reach out to each other.

Detailed Explanation/Configuration

Virtual LAN is used to separate broadcast domain in a switched environment. The reason behind separating broadcast domain can be for various reasons. Controlling ARP broadcast, unknown unicast flooding and providing security between networks are the key objectives of separating VLANs.

There are two types of trunk configuration, 1) DTP (Dynamic Trunking Protocol) and Static. For DTP, there are two key modes, Desirable and Auto where Desirable is needed to be configured in either side for the successful establishment of the trunk. The desirable mode always initiates the trunk negotiation while the auto only responds to the received initiation requests. If both sites are set as Auto, no trunk will be established.

Static trunk configuration can be done by using “switchport mode trunk” interface specific command. Let’s go to both the switches and create required VLANs, and pass them via trunk port.

Sw1 and Sw2

Go to the configuration mode and set the hostname.

```
Switch>en
Switch#configure terminal
Switch(config)#hostname SwX(X is switch number, use 1 for switch 1 and 2 for Switch 2)
```

Create VLAN 10 and 20.

```
Sw1(config)#vlan 10
Sw1(config-vlan)#exit
Sw1(config)#vlan 20
Sw1(config-vlan)#exit
```

Change the mode of the interface as an access port and assign respective VLAN ids.

```
Sw1(config)#inter fa0/2
Sw1(config-if)#switchport mode access
Sw1(config-if)#switchport access vlan 10
Sw1(config-if)#exit
Sw1(config)#inter fa0/3
Sw1(config-if)#switchport mode access
Sw1(config-if)#switchport access vlan 20
Sw1(config-if)#exit
```

Configure static trunk on the port connected to the remote switch and pass the required VLANs.

```
Sw1(config)#inter fa0/1
Sw1(config-if)#switchport mode trunk
Sw1(config-if)#switchport trunk allowed-vlan 10,20
```

Note: In order to set the IP addresses on the PCs, refer to the Lab 1.

Verification:

In the configuration section, we discussed and configured Sw1, Sw2 and the hosts based on the task requirements. Now, we have some of the verification commands to be used on the switches. Once done, we can go to the PCs and check end-to-end connectivity using PING operation.

Sw1#show vlan

VLAN Name	Status	Ports
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```
-----
-----
1      default          active    Fa0/4, Fa0/5, Fa0/6,
Fa0/7
```

Fa0/10, Fa0/11	Fa0/8,	Fa0/9,
Fa0/14, Fa0/15	Fa0/12,	Fa0/13,
Fa0/18, Fa0/19	Fa0/16,	Fa0/17,
Fa0/22, Fa0/23	Fa0/20,	Fa0/21,
	Fa0/24,	Gig0/1,
Gig0/2		
10 VLAN0010	active	Fa0/2
20 VLAN0020	active	Fa0/3
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode
Trans1	Trans2							
1	enet	100001	1500	-	-	-	-	0
0								
10	enet	100010	1500	-	-	-	-	0
0								
20	enet	100020	1500	-	-	-	-	0
0								
1002	fddi	101002	1500	-	-	-	-	0
0								
1003	tr	101003	1500	-	-	-	-	0
0								
1004	fdnet	101004	1500	-	-	-	ieee	0
0								
1005	trnet	101005	1500	-	-	-	ibm	0
0								

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode
Trans1	Trans2							

Remote SPAN VLANs

Primary	Secondary	Type	Ports

The above highlighted portion indicates that the fa0/2 and Fa0/3 are part of VLAN 10 and VLAN 20 respectively.

Let's recall that there are few other VLAN numbers predefined on a Cisco IOS switch which are hard coded onto the operating system and cannot be deleted. VLAN 1 is considered as a management VLAN and default NATIVE VLAN for the Cisco switch.

Native VLAN

In a switched environment, a trunk port refers that all the frames that pass through a trunk port should be TAGGED with a VLAN ID, tagging VLAN number attracts other operations like CRC recalculation and adding extra dot1q overhead, which in some cases are considered as an unnecessary CPU burden. Some of the frames that are required as a control function between switches like CDP, STP BPDUs can be sent to the remote devices via trunk port UNTAGGED using NATIVE VLAN. We can change native VLAN to some other numbers as well.

Sw1#show inter trunk

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Fa0/1	10,20

Port	Vlans allowed and active in management domain
Fa0/1	10,20

Port	Vlans in spanning tree forwarding state and not pruned
Fa0/1	10,20

The trunk port has default encapsulation type as an IEEE 802.1q, aka, dot1q with the default native VLAN set as 1.

Also, the trunk port only allows VLAN 10 and 20.

Note : Follow the same procedure for Sw2 also.

Now, check the connectivity between PC1 and PC3 and PC2 to PC4.

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.44

Pinging 192.168.1.44 with 32 bytes of data:

Reply from 192.168.1.44: bytes=32 time<1ms TTL=128
Reply from 192.168.1.44: bytes=32 time<1ms TTL=128
Reply from 192.168.1.44: bytes=32 time<1ms TTL=128
Reply from 192.168.1.44: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.44:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms